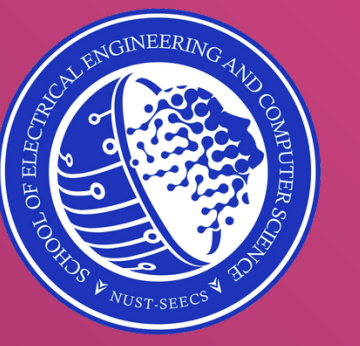




MINE-MAPPER

LIDAR GUIDED ROVER FOR LANDMINE DETECTION

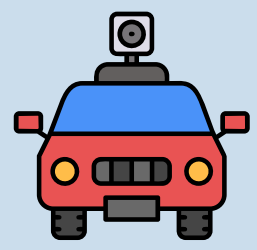
PROBLEM STATEMENT

Landmines in post-war and disaster-hit areas pose a severe threat to human life, and current detection methods are slow, risky, and costly. There is a need for an autonomous, low-cost system capable of safely detecting and mapping landmines in challenging terrains.

OBJECTIVES

- To develop an autonomous rover capable of mapping hazardous terrains using LiDAR and ROS2.
- To detect buried landmines through multi-sensor integration (metal, magnetic, and gas sensing).
- To enable autonomous navigation and obstacle avoidance in hazardous terrains.
- To provide a low-cost, efficient, and safe alternative to manual mine detection methods.

KEY FEATURES



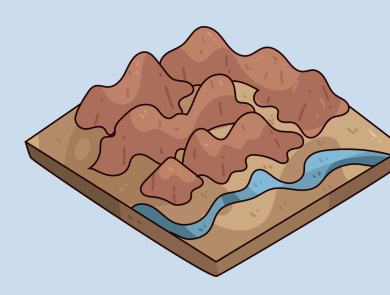
Autonomous Navigation



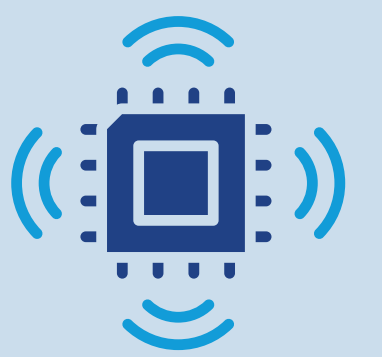
Landmine Detection



Remote Monitoring

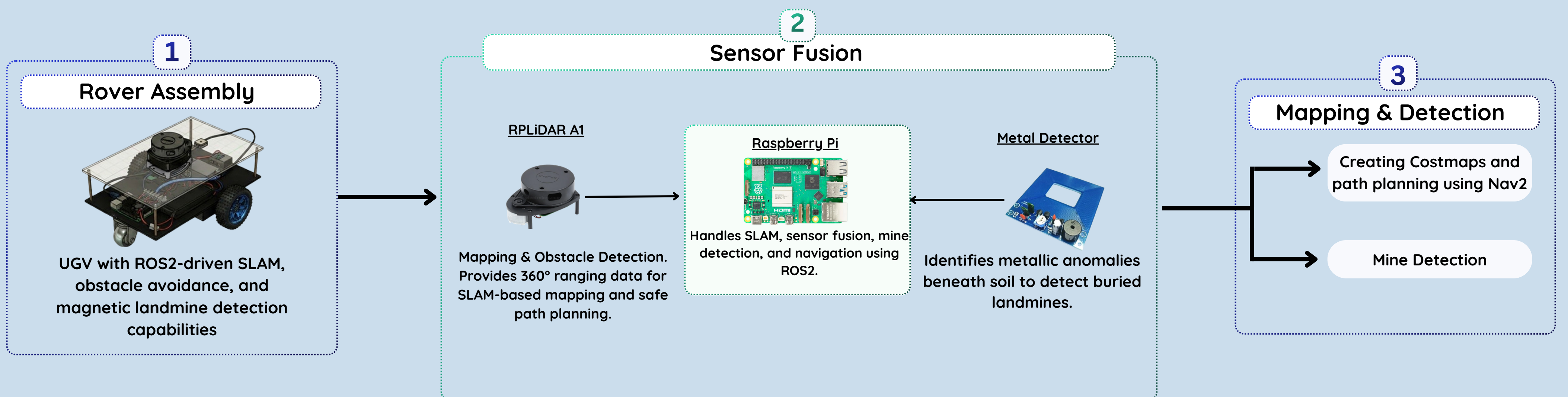


Robust Terrain Handling



Multi Sensor Fusion

METHODOLOGY



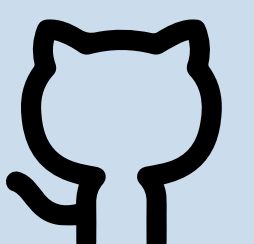
PROPOSED SOLUTION

A LiDAR-guided unmanned ground rover equipped with ROS2 is designed to autonomously map the environment using SLAM, identify mine-risk zones through sensor fusion, and navigate safely using Nav2. Detected landmine locations are identified and reported for further action, offering a scalable and affordable mine detection solution.

CONCLUSION

The developed rover demonstrates that LiDAR-based SLAM and multi-sensor fusion can provide a safe and effective way to detect and map landmines in challenging terrains. By automating navigation and detection, the system reduces risk to human personnel and provides reliable location data for future clearance operations, providing a reliable and cost-efficient system for the safe detection and localization of landmines.

ROS2



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